SOL PROTOCOL

Dungeon Level Design

Layout, flow and rules of two of the dungeons built for the project

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1 Overview

Dungeons are small, self-contained levels found in sectors. Players enter through an airlock, solve the dungeon with puzzles and movement (there is no combat), and bring the loot back to the ship. General rules, metrics and setup are in **Level Design: General Info**.

Every dungeon has a **fixed structure**: rooms, connections and solution steps never change. Some **internal sections** change from run to run, so that exploring the same dungeon again still feels different.

2 Parkour Dungeon

A hub-and-spoke dungeon built around movement. From a central room, four side appendices lead to four buttons; each button turns off one of the laser barriers that close the shaft to the loot room. Gravity is the key to the last step: it stays on for the parkour and is turned off to fly up to the loot.

2.1 At a glance

Parameter	Value
Type	Platform
Blueprint	`BP_ParkourDungeon`
Core mechanics	Platforming, gravity on/off, laser barriers linked to buttons
Main loot	3 chests in the loot room on the upper floor
Bonus loot	3 hidden chests with minor loot, one in each of the red, green and pink appendices
Randomized	The platform sections inside the appendices

2.2 Dungeon map

The numbers on the map match the steps of the dungeon flow. The loot room (top right) is drawn apart: in the dungeon it sits on the upper floor, above the gravity button.

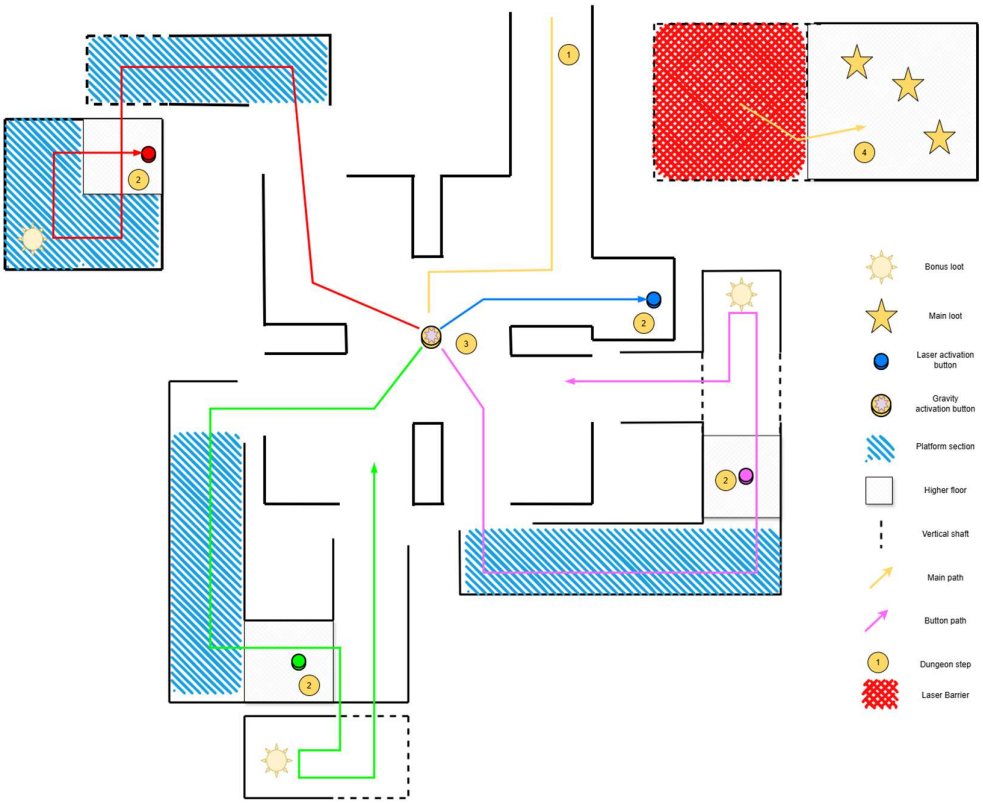


Figure 1 *Parkour Dungeon map: central room with the gravity button, the four appendices with their buttons, and the loot room on the upper floor.*

2.3 Dungeon flow

1 Enter and explore Map step 1

The player comes in through the airlock and follows the corridor to the central room. Gravity is **on**.

From the room the player can see the four side appendices and, in the middle, the **gravity button**. Above the button, a vertical shaft leads to the upper floor. The shaft is closed by **four laser barriers**, each linked to one of the buttons in the appendices.

2 Press the four buttons Map step 2

Each appendix ends with a laser button. The **blue appendix** comes first: a short corridor with no parkour, it shows the player that the barriers are turned off by buttons, so they go looking for the others. The other three (red, green, pink) are reached through a **platform section** that leads up to a higher floor.

Each button turns off the laser barrier of the same color, so the player sees the shaft opening one barrier at a time. The three main appendices also hide a bonus chest off the route (see **Appendices**).

3 Fly up the shaft Map step 3

With all four barriers off, the player goes back to the central room and presses the **gravity button** to turn gravity **off**, then flies up the shaft to the upper floor.

4 Main loot and exit Map step 4

The upper floor holds the loot room with the **3 main loot chests**. The player takes the loot and goes back to the airlock to leave the dungeon.

2.4 Appendices

Appendix	Position	Route to the button	Bonus loot	Way back
Blue	East	Short corridor with no platform section: the button is at the end.	None	Same way back
Red	North-west	Platform section along the north side, then an L-shaped platform area that ends on a higher floor.	In the corner of the L-shaped platform area.	Same way back
Green	South-west	Long platform section heading south, ending on a higher floor.	In a room below the button, through a vertical shaft.	Separate corridor north
Pink	South-east	Platform section along the south side, then a higher floor.	At the top of a vertical shaft above the button.	Separate corridor west

2.5 Co-op rule

In co-op, one player could turn gravity off while the others are still in the appendices, letting them fly over the platforms. To prevent this, when gravity is turned off:

- the appendices close with laser barriers;
- inside each appendix, a barrier sweeps the shaft from top to bottom, clearing it of players.

> **Why the shaft is the goal**
> The player sees the closed shaft and its four barriers from the first room, so the goal is clear from the start. Every button turns off a barrier the player can see, which gives clear feedback on progress.

2.6 In-engine result



Figure 2 *Top-down view. The airlock is at the top; the red grid is the laser barrier in the shaft above the central room.*

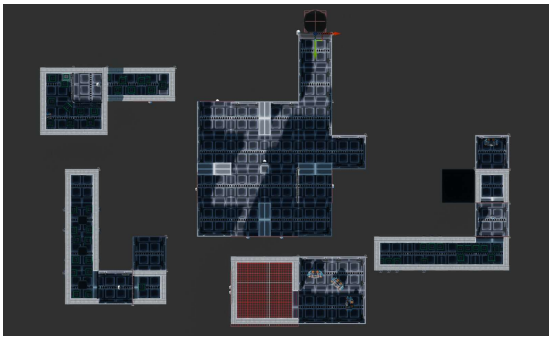


Figure 3 *Exploded top-down view: the central room and the four appendices are pulled apart, and the upper floor (laser barrier and loot room with the three chests) is shown on its own, bottom center.*

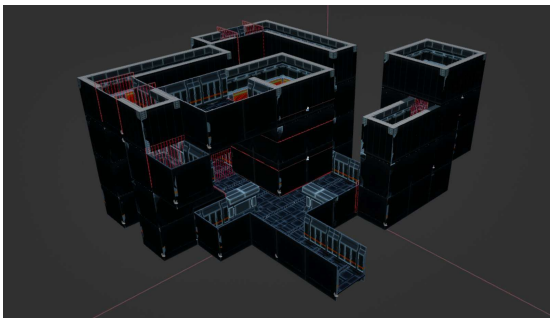


Figure 4 *Side view, with the entrance corridor in the foreground.*

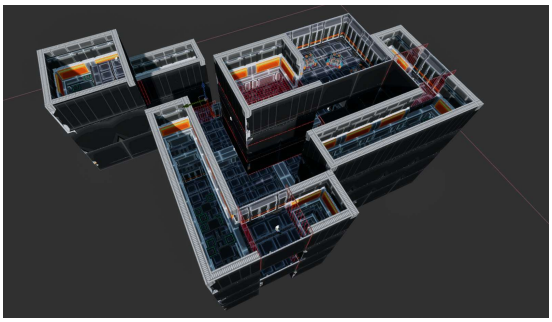


Figure 5 *Side view from above, with the loot room at the top of the central block.*

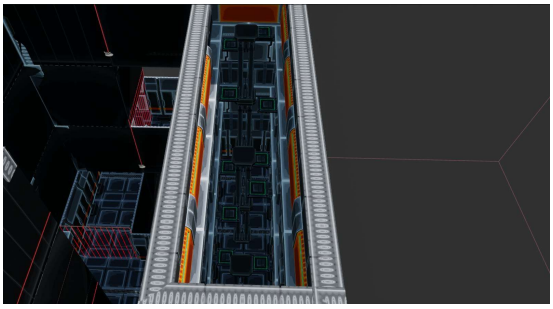


Figure 6 *Platform section in a long appendix, seen from above.*

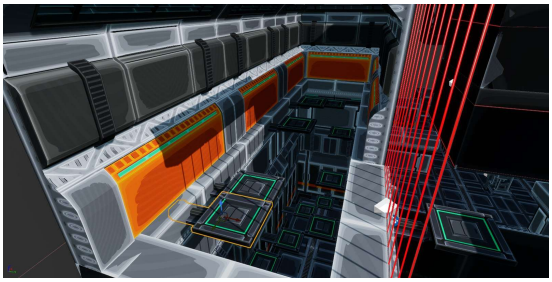


Figure 7 *Platform section with a higher floor, next to a laser barrier.*

3 Gravity Dungeon

A dungeon that tests how well the player moves in zero gravity. The player squeezes through the gaps between lasers, rolling to find the right angle, without touching them. The loot room is visible from the start through large windows, but the way in leads through two laser areas.

3.1 At a glance

Parameter	Value
Type	Gravity
Blueprint	`BP_EX_GravityDungeon3`
Core mechanics	Zero-gravity movement (roll and pitch to fit through gaps), laser obstacles, laser walls, doors linked to buttons
Main loot	2 chests in the loot room
Bonus loot	2 hidden chests: one in the vertical shaft above the upper corridor, one next to the green button
Randomized	The laser barriers and obstacles

3.2 Dungeon map

The numbers on the map match the steps of the dungeon flow. Each door has the same color as the button that opens it.

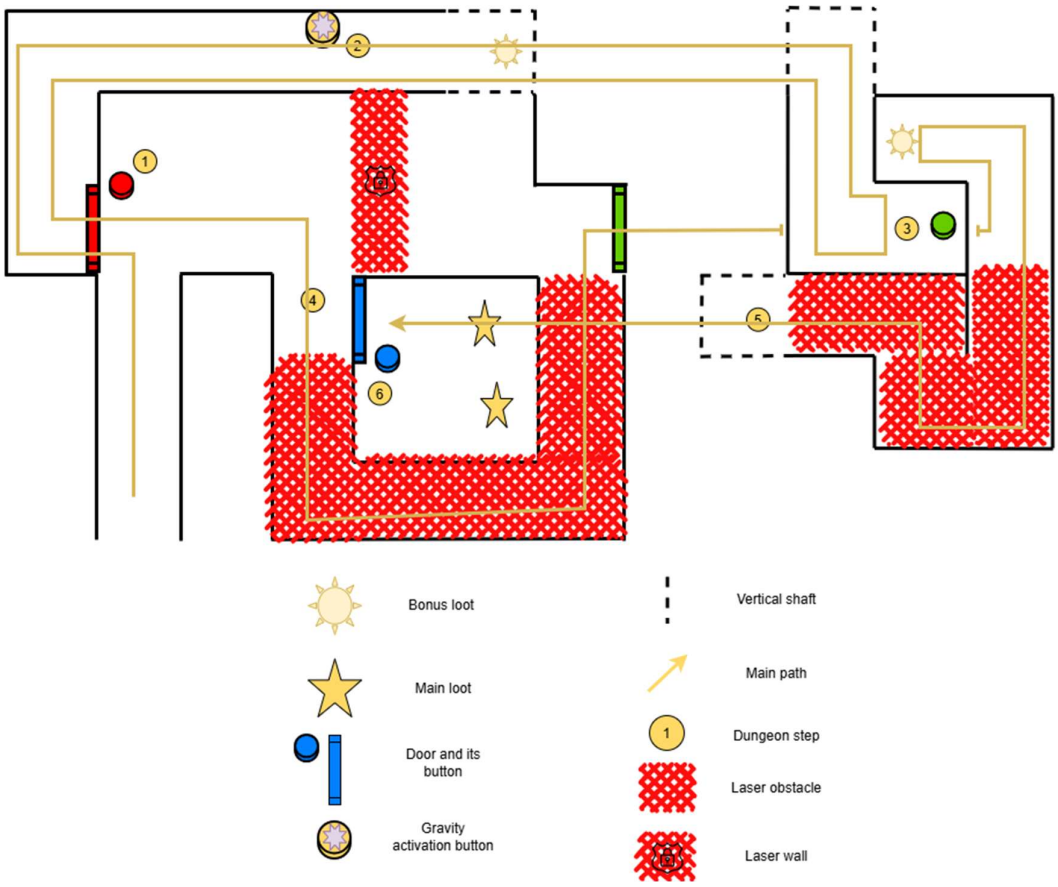


Figure 8 *Gravity Dungeon map: entrance and red door, upper corridor with the gravity button, the green button at the end of the vertical shaft, the main room with the laser obstacles, the laser corridor and the loot room.*

3.3 Dungeon flow

1 Enter Map step 1

The player comes in through the airlock and opens the **red door** with its button.

2 Turn gravity off Map step 2

In the corridor beyond the door, the player presses the **gravity button** to turn gravity **off**. From here on, the whole dungeon is played in zero gravity.

3 Up the shaft to the green button Map step 3

The player finds the entrance to the **vertical shaft** high up, flies through it to the end and presses the **green button**. The button opens the second door of the main room (not the loot room door) and turns on the lasers.

The button is right above that door and the floor is a grating, so when the player presses it they hear the sound and see the door open on the floor below. A bonus chest is hidden in the shaft above the upper corridor, and another one next to the button.

4 Cross the main room Map step 4

Back in the main room, the player finds it full of **laser obstacles**. The short route is blocked by a **laser wall**, so they take the long route around the loot room to the newly opened **green door**.

5 Laser corridor Map step 5

Behind the green door, another vertical shaft leads to a corridor full of lasers, where the walls are lasers too. Movements here must be very precise. At the end of the corridor, the player drops into the loot room from above.

6 Loot room and exit Map step 6

The player takes the **2 main loot chests** and opens the **blue door** with its button. Opening the door turns off all the lasers, and the player is free to go back to the ship with the loot.

> **Loot in sight, lasers out of sight**
> When the player enters, most lasers are still off, so they can explore the main room calmly. The loot is clearly visible through large windows but cannot be reached. Pressing the green button turns the room they already know into a laser course.

3.4 In-engine result

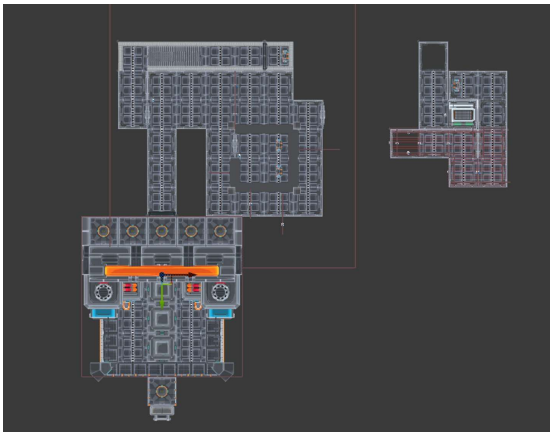


Figure 9 *Top-down view. On the right, the second floor of the dungeon, with the passage to the loot room.*

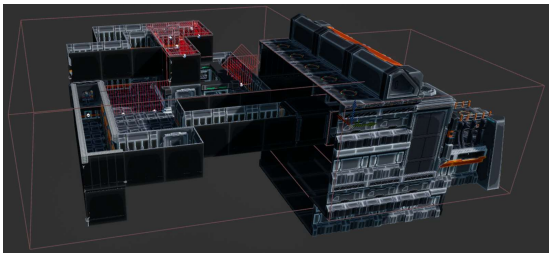


Figure 10 *Side view. The laser areas are shown in red.*